

# Skills Summary

## Using Slopes to Prove Lines Parallel or Perpendicular

Use this reference while completing your worksheet or when studying.

### Skill 1 — Parallel Lines Have Equal Slopes

**Slope formula:** for points  $(x_1, y_1)$  and  $(x_2, y_2)$ ,  $m = \frac{y_2 - y_1}{x_2 - x_1}$  — the rise over the run.

**Parallel test:** two distinct non-vertical lines are parallel *exactly when*  $m_1 = m_2$ . Equal slopes mean equal direction, so the lines never meet. Two vertical lines are parallel as well, even though neither has a slope.

**Example:** Line  $\ell$  passes through  $A(1, 1)$  and  $B(5, 3)$ . What is the slope of every line parallel to  $\ell$ ?

**Step 1.** Parallel lines carry the same slope, so the whole question reduces to finding the slope of  $\ell$  itself.

**Step 2.**  $m = \frac{3 - 1}{5 - 1} = \frac{2}{4}$ , reduced to lowest terms.  
 $\Rightarrow m = \frac{1}{2}$

**Try it:** Line  $n$  passes through  $C(-2, -1)$  and  $D(2, 5)$ . What is the slope of every line parallel to  $n$ ?

check:  $\frac{2}{2} = 1$

### Skill 2 — Slopes of Perpendicular Lines

**Perpendicular test:** two non-vertical lines are perpendicular *exactly when*  $m_1 m_2 = -1$ ; equivalently, each slope is the **opposite reciprocal** of the other — flip the fraction and change the sign.

**The exception:** a vertical line has no slope at all, so the product rule cannot be used. A line perpendicular to a vertical line is horizontal, and its slope is 0.

**Example:** Find the slope of a line perpendicular to the line through  $E(-1, 2)$  and  $F(3, 4)$ .

**Step 1.** Perpendicular means opposite reciprocal, so find the slope of  $\overline{EF}$  first and then transform it.

**Step 2.**  $m_{EF} = \frac{4 - 2}{3 - (-1)} = \frac{2}{4}$ , which is  $m_{EF} = \frac{1}{2}$ ; flipping it gives 2 and changing the sign gives the perpendicular slope. Check:  $(\frac{1}{2})(-2) = -1$  ✓  
 $\Rightarrow m = -2$

**Try it:** Find the slope of a line perpendicular to the line through  $G(0, 5)$  and  $H(2, -1)$ .

check:  $\frac{2}{1} = 2$  is the perpendicular slope, so  $m = -\frac{1}{2}$

**(!) Watch out:** the opposite reciprocal changes *both* things. The perpendicular partner of  $\frac{2}{3}$  is  $-\frac{3}{2}$  — not  $\frac{3}{2}$  (sign forgotten) and not  $-\frac{2}{3}$  (fraction not flipped).

## Skill 3 — Coordinate Proofs Using Slopes

**How a coordinate proof works.** Turn the claim into slopes, compute, compare, then say what the comparison proves:

- both pairs of opposite sides parallel  $\Rightarrow$  parallelogram
- parallelogram with one pair of adjacent sides perpendicular  $\Rightarrow$  rectangle
- midpoints come from  $(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$ ; lengths from the distance formula

Finish with a sentence naming the theorem you used — the arithmetic alone is not a proof.

**Example:** A triangle has third side  $\overline{TU}$  from  $T(0, 0)$  to  $U(8, 4)$ , and the midpoints of the other two sides are  $M(1, 3)$  and  $N(5, 5)$ . Show  $\overline{MN} \parallel \overline{TU}$ .

**Step 1.** Parallel is a statement about direction, so compare the slope of the midsegment with the slope of the third side.

**Step 2.**  $m_{MN} = \frac{5-3}{5-1} = \frac{2}{4}$ , and  $m_{TU} = \frac{4-0}{8-0} = \frac{4}{8}$  — the same number, so the segments are parallel.

$$\Rightarrow m = \frac{1}{2}$$

**Try it:** The midpoints of two sides of a triangle are  $(2, 4)$  and  $(7, 6)$ . Find the slope of the midsegment joining them.

$$\frac{5}{2} = m \text{ :check}$$